

1. Administrative & Study Identification

Full Study Title: A Multicentre, Single-Arm, Phase 3b Study to Assess Changes in Symptoms in Adult Participants With Chronic Rhinosinusitis With Nasal Polyposis Initiating Treatment With Tezepelumab **Short Title/Acronym:** ESSENCE **Protocol Number:** D5242C00002 **NCT Number:** NCT06706817 **Sponsor Name:** AstraZeneca **Investigational Product:** Tezepelumab (TEZSPIRE®) **Phase of Development:** Phase IIIb **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate treatment outcomes by describing changes from baseline in participant-reported nasal congestion as evaluated by the nasal congestion score (NCS) and sino-nasal symptoms evaluated by the 22-item Sino-Nasal Outcome Test (SNOT-22) following the initiation of tezepelumab. **Secondary Objectives:** To determine the proportion of SNOT-22 responders, the time to first response for SNOT-22, the time to first response for Total Symptom Score (TSS), and to evaluate endoscopic findings for histological changes and Nasal Polyp Score (NPS) control from baseline to Week 24.

2.2 Background & Rationale Chronic Rhinosinusitis with Nasal Polyposis (CRSwNP) can be a severe, debilitating condition characterized by persistent nasal congestion, loss of smell, and sleep disturbances. Tezepelumab is a monoclonal antibody targeting the alarmin thymic stromal lymphopoietin (TSLP). This study aims to provide real-world-like evidence on symptom relief and treatment outcomes for adult participants with physician-determined, surgery-eligible CRSwNP whose symptoms remain uncontrolled despite background intranasal corticosteroid therapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 180 participants **Number of Sites:** Approximately 60 sites across 10 countries (Europe, US, Canada) **Estimated Study Start:** 03-Dec-2024 **Estimated Primary Completion:** 13-Jan-2027 **Estimated Study Completion:** 13-Jan-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female ≥ 18 years of age. 2. Physician-diagnosed CRSwNP for at least 12 months prior to Visit 1 with severity consistent with the need for surgery (total NPS ≥ 4 , with at least 2 for each nostril, confirmed by a central reader). 3. SNOT-22 total score ≥ 30 and mean NCS ≥ 2 during the 2 weeks prior to Visit 2, with ongoing documented NP symptoms for > 8 weeks. 4. Receiving standard of care for CRSwNP, including a stable dose of intranasal corticosteroids (INCS) for at least 30 days prior to Visit 1.

4.2 Exclusion Criteria 1. Sinus surgery within 6 months of the screening visit OR any past sinus surgery that changed the lateral wall of the nose making NPS evaluation impossible. 2. Documented allergic fungal rhinosinusitis and/or central compartment atopic disease. 3. Any clinically important pulmonary disease other than asthma (e.g., active lung infection, cystic fibrosis, bronchiectasis) that could confound interpretation of endpoints. 4. Receipt of any marketed or investigational biologic agent within 4 months prior to Visit 1.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Tezepelumab 210 mg single dose administered once every 4 weeks (Q4W). **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** Up to 24 weeks of active treatment (Total study duration up to 40 weeks, including screening and safety follow-up).

5.2 Concomitant & Prohibited Therapy Permitted: Stable background dose of intranasal corticosteroids (INCS) and asthma maintenance therapy if applicable. **Prohibited:** Other biological therapies, systemic immunosuppressants, or investigational agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Weeks -4 to 0	Informed Consent, Medical History, SNOT-22 scoring, Endoscopy (central reader NPS)
Baseline	Week 0	First Tezepelumab Dose, baseline daily symptom diary review
Interim	Weeks 4, 8, 12, 16, 20	Q4W IP Administration, Safety Labs, Efficacy Assessment (NCS, SNOT-22, Endoscopy at specific intervals)
End of Study	Week 24 & Week 36	End of Treatment (W24), IP Accountability, Safety Follow-Up (W36), Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change from baseline in participant-reported nasal congestion as evaluated by the Nasal Congestion Score (NCS) and change from baseline in sino-nasal symptoms as evaluated by the SNOT-22 questionnaire at Week 24. **Measurement Method:** Electronic daily symptom diaries and clinical questionnaires.

7.2 Secondary & Exploratory Endpoints 1. Proportion of SNOT-22 responders (Minimal Clinically Important Difference [MCID] from baseline = -8.9) at each collected timepoint. 2. Time to first response for SNOT-22, time to first response for Total Symptom Score (TSS), and time to first response for NP control. 3. Endoscopy findings describing histological changes and CRSwNP visualization from baseline to Week 24.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection via investigator assessment and patient reports at each Q4W site visit, extending through the 12-week safety follow-up period (Week 36). **Serious Adverse Events (SAEs):** Standard reporting timeline within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal safety monitoring managed by AstraZeneca.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Full Analysis Set / Intent-to-Treat (ITT) population for primary outcomes. Safety Set for all AE/SAE evaluations. **Sample Size Justification:** $N = 180$ participants (including a pre-specified ~50 participants with a baseline NPS = 4), appropriately powered to describe symptom trajectories and response rates. **Handling of Missing Data:** Analysis of covariance and mixed models for repeated measures (MMRM) analyzing data at all collected timepoints to account for intermittent missing daily diary entries.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits; centralized blinded review of endoscopic Nasal Polyp Scores (NPS) to ensure diagnostic stringency. **Audit Trail:** eCRF locking and compliance verification of patient electronic daily symptom diaries.

11. Study Identifiers & Governance

Primary ID: D5242C00002 **Registry Number:** NCT06706817 **CTIS Identifier:** 2024-513862-20-00 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** ESSENCE **Official Regulatory Title:** A Multicentre, Single-Arm, Phase 3b Study to Assess Changes in Symptoms in Adult Participants With Chronic Rhinosinusitis With Nasal Polyposis Initiating Treatment With Tezepelumab

12. Clinical Framework (CRSwNP Specific)

Primary Condition: Chronic Rhinosinusitis with Nasal Polyposis (CRSwNP) **Diagnosis Threshold:** Total NPS ≥ 4 (at least 2 per nostril), SNOT-22 ≥ 30 , and mean NCS ≥ 2 **Medical Methodology:** Biologic therapy complementing Guideline Determined Medical Therapy (INCS) **Optimization Target:** Symptomatic improvement (NCS & SNOT-22 reduction) and surgical avoidance

13. Study Procedures & Methodology

Management Pathway: Tezepelumab Q4W alongside standard background INCS pathway **Education Protocol (HCP):** Site investigator training on endoscopic imaging requirements and central reader submission protocols **Education Protocol (Patient):** Training on completing daily symptom diaries (NCS) and SNOT-22 instruments **Quality Audit Mechanism:** Centralized endoscopic reviewer tracking & electronic diary adherence monitoring

14. Observation & Follow-Up Schedule

Total Duration: 40 Weeks (Up to 4 weeks screening, 24 weeks treatment, 12 weeks follow-up) **Onsite Visit Cadence:** Weeks 0, 4, 8, 12, 16, 20, 24, and 36 **Remote Monitoring Frequency:** Continuous via daily symptom diaries **Checklist Review Interval:** Every 4 weeks (Q4W) corresponding to clinic visits

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				